# Mark Raymond Takla

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Dual American-Egyptian Nationality | Military Exempted

## TECHNICAL SKILLS AND LANGUAGES

**Languages:** Java, Python, C, C#, SQL, MIPS Assembly, Haskell, Prolog, R, HTML/CSS.
**Frameworks & Libraries:** Next.js, React, ASP.NET, FastAPI, LangChain, LangGraph, Scikit-Learn, TensorFlow, Keras, Pandas, NumPy.
**Software & Tools:** UIPath (RPA), PowerBI, MATLAB, OrCAD Pspice, Adobe Photoshop, AfterEffects, Microsoft Office.
**Platforms:** Linux, Git, SQL Server, MongoDB, SQLite, AutoCAD.
**Spoken Languages:** Arabic & English (Proficient), German (A2), French (Basics).

## EDUCATION

**German University in Cairo (GUC)**
Bachelor of Engineering | Major in Computer Science — 09/2023 - Current
Cumulative German Scale GPA: 2.01 (B+) (≈3.5 American Scale GPA)

**Cairo English School (CES) and New Ramses College (NRC) (IGCSE System)** — Earlier Years
Early graduation after year 11 | Final score 110% with scale factor

## WORK EXPERIENCE

**German University in Cairo**
Computer Science Junior Teaching Assistant — 09/2024 – 06/2026
- CSEN102/CSIS102 (Python): Instructed **60+** students in core programming and data structures.
- CSEN401 (OOP Java): Mentored **50+** students and held 2 weekly project office hours to guide complex software projects.
- Gained many soft-skills such as leadership, communication, teaching, patience, accountability, and time management.

**TPAY**
Integration Engineer Intern — 08/2026 – 09/2026
- Cut manual testing time by **85%** by building a production-ready AI agent tool using LangGraph to solve an 8-year-old team bottleneck.
- Integrated TPAY into merchant web and mobile platforms across 4 methods (REST APIs, Client/Server SDKs, Android SDK) using JavaScript and Postman. (Demo Link)

**Bank NXT**
Robotic Process Automation (RPA) Internship — 07/2025 – 08/2025
- Completed a multidisciplinary rotation across the banking industry, including intensive weeks in RPA, Integration and IT Governance & Risk Management.
- Developed **5+** unattended UIPath robots for automated data entry and web scraping, saving significant bank resources and eliminating manual form filling time by 100%. (Demo Link)

**GUC Innovators and GUC Google Developer Groups**
Vehicle Dynamics Engineer in Innovators and Robotics Engineer in GUC GDG — 06/2024 – 07/2025

**Alstom**
Rotational Engineering Internship — 06/2024 – 08/2024
- Completed a comprehensive rotation through V&V, ICC, CBI, and RCM departments, applying SCADA systems and ISTQB principles to ensure (ATC/ATS) mission-critical reliability.
- Identified **4** technical discrepancies in signaling test cases within the HMI to ensure accurate train simulations for the Turkey UAG line.

## PROJECTS (*Full List (25+) on GitHub and Portfolio Website*)

**University HR Management System (GitHub Repository)** — 12/2025
- Engineered a comprehensive relational database using SQL Server DBMS (including EERD and relational schema) with **40+** stored procedures to drive a responsive ASP.NET & JS fully functional website for multi-rank data management and automated payroll logic.

**Customer churn benchmarking on two tabular datasets (GitHub Repository)** — 03/2026
- Benchmarked Random Forests and decision trees against tuned Logistic Regression using Scikit-learn pipelines and GridSearchCV.

**Production-Ready Email AI Assistant (GitHub Repository)** — 04/2026
- Built an automated email assistant using LangGraph Cloud, featuring a React-based Chat UI that supports Human-in-the-loop interrupts and persistent thread management.

**Jackaroo Multiplayer Computer Game with GUI (Java) (GitHub Repository)** — 05/2025

**Advanced Persistent RDBMS Engine (GitHub Repository)** — 05/2026
- Engineered a high-performance engine using a 3-tier hierarchy (DBMS, Table, Page) with disk-based serialization and O(1) pointer-based record access.

## Certificates, Competitions and Volunteering (Full list of 11+ Certificates on Portfolio Website)

- **Certifications**: *Data Science & AI Diploma* (Sprints.ai - 250hrs and Space Code 180hrs); *Intro to LangChain* (LCA); *AI Data Analytics Bootcamp* (EYouth); *Data Analysis Using R* (SRTA-City).
- **Community**: Scout Leader (2013–Present); Managed summer camps in New Jersey, USA for 30+ children; Organized charity activities and ceremonies for 200+ Sudanese-Egyptian refugee children.